

Week 7 XPCyber Challenge Networking Anomalies: A hunt for the hidden

Week #7 XPCyber Challenge - Networking Anomalies: A hunt for the hidden

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Term 2

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Executive Summary:

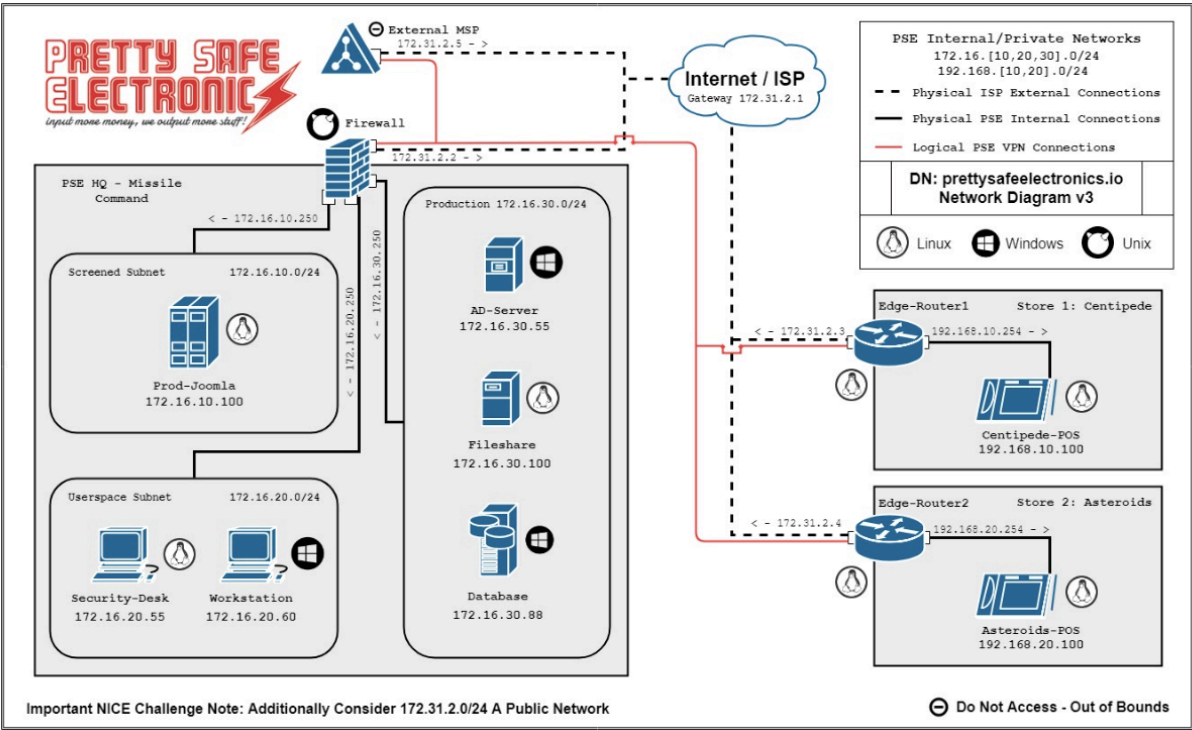
In this lab one of our coworkers had three packet capture files they needed us to look through for malicious activity. We were asked to comb through the files, locate and document the packet range of the suspicious activity. Due to the large number of frames we used the conversations tab in the wireshark menu and filters to locate IP addresses from outside of our network. After finding the frame range of suspicious activity we would submit the information to a company webform, listing the packet range of the activity and its associated IP address. We first searched for the "Prod-Joomla" Packet Capture. Where one of the IP addresses was outside of our network –172.31.2.153– we confirmed by looking at the network map as the external msp address started with 172.31. We applied it as a filter, noting almost 200 TCP traffic frames from the IP starting at frame 8527 and going until frame 8703. We followed the same steps for the second and third packet. For the second packet “workstation” IP 172.31.2.153 showed up again, starting at frame 9142 ending at frame 12779. For the final packet capture "Fileshare" the relevant TCP traffic was included in packets 141 to 2488 with the same IP address. Once finding out the network IP and the packet range of the potentially malicious activity we filled out the formbot and concluded our investigation. One likely cause of the unusual TCP traffic is unauthorized access or command and control traffic with automated activity this can be seen through extended packet ranges. Our first recommendation is to investigate the external IP address and block it at a firewall level (Scarfone & Hoffman, 2009). This prevents additional communication from the IP to the network. The second recommendation would be a full security review of all systems that the suspicious IP connected with (SI-4: Information System Monitoring - CSF Tools, 2020). Once done, we recommend configuring the IDS to alert the

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administrator of unusually large TCP traffic and then to implement stricter outbound traffic rules to help prevent internal systems from connecting with external IPs (Nelson et al., 2025).

Technical Steps & Analysis

Network map



Environmental Chat Logs

Ashley Steele

I have some packet capture files I need someone to comb through for any type of malicious activity that might be happening. Unfortunately, I'm not very familiar with analyzing pcap files. My skill set is limited to generating them from capture traffic.

Ricardo Cortes

I'm pretty positive that @playerone can be of assistance here.

Ashley Steele

That would be greatly appreciated, @playerone. I suspect you might find some evidence of malicious activity occurring on our network, so please exercise extreme scrutiny.

Ashley Steele

I put the pcap files I need you to look through on Security-Desk located at /home/playerone/Desktop. Please fill out the Incident Response Form for each of the files, regardless of if you find anything or not.

Jacques Raffin

Ok that has me a bit worried, if you do find something make sure you fill out the web form with the most accurate detail that you can. There is a field asking for the packet range of the malicious activity. We're specifically looking for the first and last packet considered to be part of the event in question. If you don't find anything in a specific pcap file just answer 'none' in the fields that ask for malicious IP address and packet range.

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Notes

- Looking for malicious activity
- 3 files, limited information
- Report any unusual or suspicious traffic using online form
- Different IP for different team members

Thought Process:

We started by logging into our security desk machine and located the packet capture files on the desktop. There were 3 files total with the names “workstation.pcap”, “Prodjoomla.pcap” and “fileshare.pcap” –Figure 1–

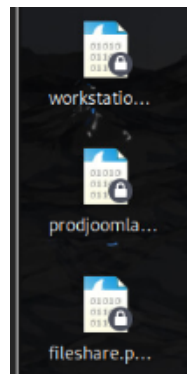
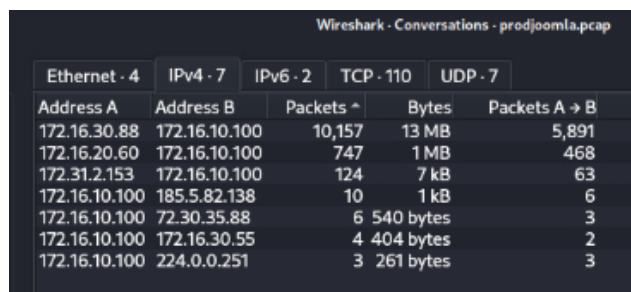


Figure 1

We began our investigation with the "prodjoomla" Packet Capture. At first glance nothing appeared unusual so we looked deeper into the packet using the conversations tab under statistics. Under conversations we sorted it by the number of packets to see who was conducting larger parts of the network traffic. In the conversations tab we noticed that one of the IP

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addresses was outside of our network based on the network diagram. This IP was suspicious and out of network with an IP address of 172.31.2.153. – Figure 2–

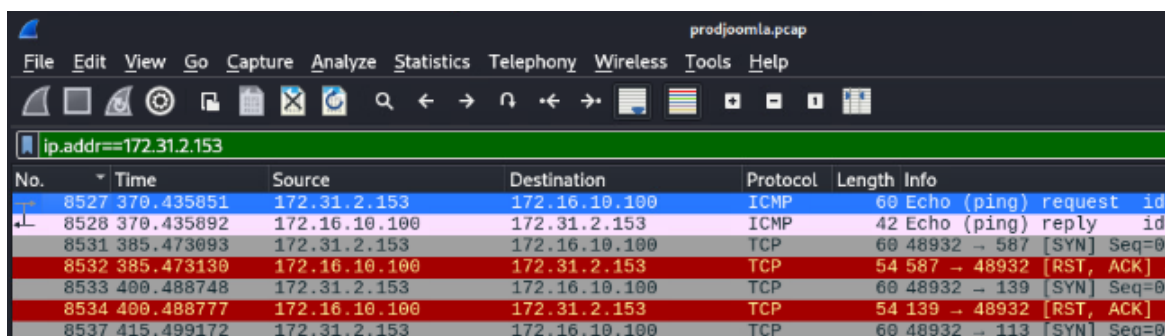


Wireshark · Conversations · prodjoomla.pcap

Ethernet · 4	IPv4 · 7	IPv6 · 2	TCP · 110	UDP · 7
Address A	Address B	Packets ↕	Bytes	Packets A → B
172.16.30.88	172.16.10.100	10,157	13 MB	5,891
172.16.20.60	172.16.10.100	747	1 MB	468
172.31.2.153	172.16.10.100	124	7 kB	63
172.16.10.100	185.5.82.138	10	1 kB	6
172.16.10.100	72.30.35.88	6	540 bytes	3
172.16.10.100	172.16.30.55	4	404 bytes	2
172.16.10.100	224.0.0.251	3	261 bytes	3

Figure 2

We then applied that IP address as a filter for the packet capture. – ip.addr== 172.31.2.153– . We noticed around 176 frames of TCP traffic in the packets. After scrolling through we noted that the suspicious traffic from the IP started at packet 8527 and continued until packet 8703. –Figure 3– Based on the patterns present we found what appears to be TCP port scanning, TCP port scanning is when you “send a SYN packet to all 65536 ports and record all SYN/ACK's coming back and conclude the ports which are opened based on the reply” (Cyber Security Network Mapping & Port Scanning, n.d.). Since it is an out of network IP scanning all of our port It could be an attacker trying to find open ports to exploit on our network meaning that we must carefully document all of the information. to find solutions to prevent this.



prodjoomla.pcap

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

ip.addr==172.31.2.153

No.	Time	Source	Destination	Protocol	Length	Info
8527	370.435851	172.31.2.153	172.16.10.100	ICMP	60	Echo (ping) request id=
8528	370.435892	172.16.10.100	172.31.2.153	ICMP	42	Echo (ping) reply id=
8531	385.473093	172.31.2.153	172.16.10.100	TCP	60	48932 → 587 [SYN] Seq=0
8532	385.473130	172.16.10.100	172.31.2.153	TCP	54	587 → 48932 [RST, ACK] S
8533	400.488748	172.31.2.153	172.16.10.100	TCP	60	48932 → 139 [SYN] Seq=0
8534	400.488777	172.16.10.100	172.31.2.153	TCP	54	139 → 48932 [RST, ACK] S
8537	415.499172	172.31.2.153	172.16.10.100	TCP	60	48932 → 113 [SYN] Seq=0

Figure 3

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We filled out the online form with the relevant information and moved onto the second packet.

The second packet was the "workstation" packet. We followed the same steps here and found the same IP address with large amounts of TCP traffic. We again selected statistics and conversations. We noted that the same suspicious out of network IP address was 172.31.2.153.

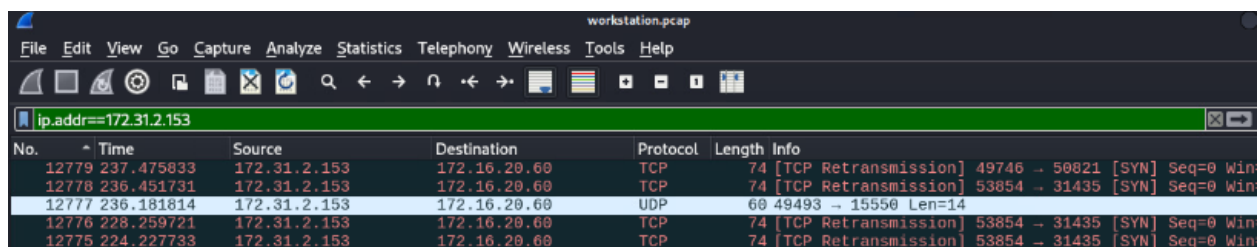
–Figure 4–

Wireshark - Conversations - workstation.pcap

Ethernet - 6	IPv4 - 39	IPv6 - 1	TCP - 2162	UDP - 54	
Address A	Address B	Packets ^	Bytes	Packets A → B	Bytes A → B
172.16.20.60	8.253.197.121	4,314	4 MB	1,264	70 kB
172.31.2.153	172.16.20.60	2,441	173 kB	2,282	147 kB
172.16.20.60	64.4.54.18	2,037	2 MB	666	206 kB

Figure 4

We then applied that IP address as a filter for the packet capture. In this packet capture the suspicious traffic ranged from packet 9142 to packet 12779. –Figure 5– There was a significantly higher number of frames captured in this packet relating to the suspicious out of network IP. This information was uploaded into the online form. We can see in this packet that there was repeated attempts to make contact with one of our devices in network specifically our workstation machine but they continually failed.



No.	Time	Source	Destination	Protocol	Length	Info
12779	237.475833	172.31.2.153	172.16.20.60	TCP	74	[TCP Retransmission] 49746 → 50821 [SYN] Seq=0 Win
12778	236.451731	172.31.2.153	172.16.20.60	TCP	74	[TCP Retransmission] 53854 → 31435 [SYN] Seq=0 Win
12777	236.181814	172.31.2.153	172.16.20.60	UDP	60	49493 → 15550 Len=14
12776	228.259721	172.31.2.153	172.16.20.60	TCP	74	[TCP Retransmission] 53854 → 31435 [SYN] Seq=0 Win
12775	224.227733	172.31.2.153	172.16.20.60	TCP	74	[TCP Retransmission] 53854 → 31435 [SYN] Seq=0 Win

Figure 5

We completed our investigation with the final packet capture "Fileshare" The same out of network IP address was present in this file as well. To verify there were no other suspicious IP's

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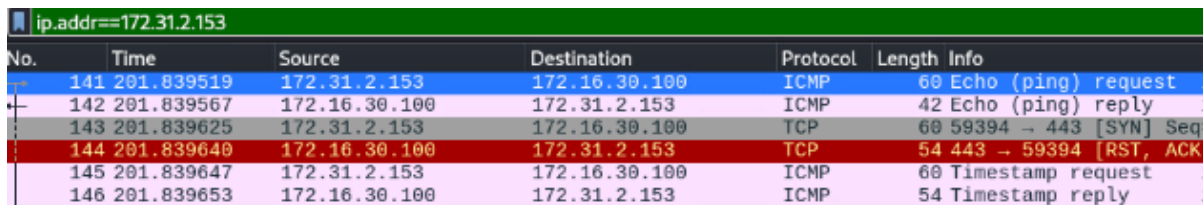
we went through the same steps of opening statistics and then conversations and checked all documented IP addresses and only found the 172.31.2.153 as out of network. –Figure 6–

Wireshark - Conversations - fileshare.pcap

Ethernet · 5	IPv4 · 4	IPv6 · 1	TCP · 1051	UDP · 7	
Address A	Address B	Packets ^	Bytes	Packets A → B	Bytes A → B
172.31.2.153	172.16.30.100	2,345	158 kB	1,190	81 kB
172.16.30.88	172.16.30.100	102	18 kB	55	10 kB
172.16.20.60	172.16.30.100	20	1 kB	10	692 bytes
172.16.30.88	172.16.30.255	3	276 bytes	3	276 bytes

Figure 6

Once we applied it as a filter the relevant TCP traffic was included in packets 141 to 2488. –figure 7– In this packet we could see that are out of network IP was probing our Fileshare device by first confirming availability via ICMP echo request/reply. It then tries to establish a TCP connection with the port 443 (HTTPS). At which point file share rejects the connection whether that be because of the firewall or just because it's closed then the out of network IP sends an ICMP timestamp request which could potentially be aiming to get more information about the host. Some of the information that they can gather is the “date that is set on the targeted machine, which may assist an unauthenticated, remote attacker in defeating time-based authentication protocols.” (ICMP Timestamp Request Remote Date Disclosure, n.d.) We filled out the final online form submission and concluded our physical investigation.



ip.addr==172.31.2.153

No.	Time	Source	Destination	Protocol	Length	Info
141	201.839519	172.31.2.153	172.16.30.100	ICMP	60	Echo (ping) request
142	201.839567	172.16.30.100	172.31.2.153	ICMP	42	Echo (ping) reply
143	201.839625	172.31.2.153	172.16.30.100	TCP	60	59394 → 443 [SYN] Seq=
144	201.839640	172.16.30.100	172.31.2.153	TCP	54	443 → 59394 [RST, ACK]
145	201.839647	172.31.2.153	172.16.30.100	ICMP	60	Timestamp request
146	201.839653	172.16.30.100	172.31.2.153	ICMP	54	Timestamp reply

Figure 7

Conclusions and Recommended actions

The conclusion of this investigation was based on the evidence that was gathered through the analysis of the three pcap files on security-desk where we found suspicious activity from the IP 172.31.2.153. Because this IP appeared across multiple systems and generated significantly more traffic than other external connections, it was potentially indicative of unauthorized access, automated scanning, or command and control communication. This repeated pattern across multiple packet captures provides defensible evidence supporting the conclusion that the activity may represent malicious network behavior. The most pragmatic remediation would be to investigate the external IP address and block it at a firewall level (Scarfone & Hoffman, 2009) as this prevents additional communication from the IP to the network. The second recommendation would be a full security review of all systems that the suspicious IP connected with (SI-4: Information System Monitoring - CSF Tools, 2020). Once done, we recommend configuring the IDS to alert the administrator of unusually large TCP traffic and then to implement stricter outbound traffic rules to help prevent internal systems from connecting with external IPs (Nelson et al., 2025). While implementing the previously explained actions it is imperative that we continue close monitoring of all affected systems in case there was any information gathered during the TCP port scans and ICMP requests.

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Three and two debrief

For this lab one thing that went well was that we were a lot more familiar with the process based on a previous lab. In the last lab the malicious activity was different; but several of the steps involved to find the traffic were the same. Both partners were able to find the traffic independently and come together to confirm the findings. This however leads to one area of improvement. There was some level of confusion because we did not realize that the Out of network IP addresses were different. In one partner's lab the IP ended in “.153” and in the other it ended in “.148” so at first we got the form submissions incorrect since we typed in the IP for the first partner not knowing the two instances produced different malicious IPs.

A second thing that was done well in this lab was that we were able to adjust and adapt to the changes in the environment quickly. Once we realized the Ip's were different we altered our language as we walked through the process to make sure it would be accurate for both of our labs and anyone else following our report. For example regardless of what IP your instance has as malicious you can follow the same steps we wrote out just replacing “172.31.2.153” with the IP address that shows as being out of network. A third thing that went well was that even though we are now technically on spring break we were able to plan out a time to work on the lab before we left so that we had adequate time to complete the assignment while also being sure to prioritize our break once it was complete. We selected Saturday right after break started so that once it was

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done we no longer had to worry about it and could focus on spending time with our family and friends. One final area for improvement is in the conciseness of language. There were many areas where our writing ran much longer than intended and needed to be cut down significantly. We need to work on a balance of technical writing and length while also making sure that someone with no knowledge of the process could follow and replicate our steps.

Overall even though this was a new group we were able to implement what had gone well with our past partners when completing this lab. We also discovered new challenges that we will continue to work on together for future labs.

Answers

@formbot

Name of PCAP File:

Malicious IPv4 Address:

Packet Range of Malicious Activity:

 1

@formbot

Name of PCAP File:

Malicious IPv4 Address:

Packet Range of Malicious Activity:

 1

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@formbot

Name of PCAP File:

Malicious IPv4 Address:

Packet Range of Malicious Activity:

☒ 1

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NICE Challenge Project

Challenge Submission Report

<https://portal.nice-challenge.com/reports/verify/35030-61A4-91620/>

Submission ID: 151847

Timestamp: 3/7/2026 5:11 PM UTC

Name: Anna Fanco

Challenge ID: 85

Challenge Title: Networking Anomalies: A Hunt For The Hidden



This report has not been published by a curator. The NICE Challenge Project cannot vouch for its accuracy.

Scenario

You are provided a network traffic capture file that needs to be reviewed. Investigate the current state of the network, reporting any unusual or suspicious traffic.

Duration

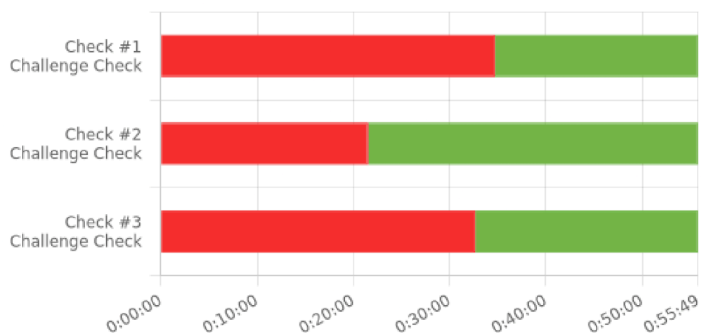
0:55

Final Check Details

- ✓ Check #1: Prod-Joomla Packet Capture Correctly Analyzed
- ✓ Check #2: Workstation Packet Capture Correctly Analyzed
- ✓ Check #3: Fileshare Packet Capture Correctly Analyzed

Full Check Pass

Full: 3/3



Element

Cybersecurity

Work Role

Cyber Defense Analyst

DCWF Tasks

- 433: Characterize and analyze network traffic to identify anomalous activity and potential threats to network resources.

Knowledge, Skills, and Abilities

- 22: Knowledge of computer networking concepts and protocols, and network security methodologies.
- 81A: Knowledge of network protocols such as TCP/IP, Dynamic Host Configuration, Domain Name System (DNS), and directory services.
- 87: Knowledge of network traffic analysis methods.
- 92: Knowledge of how traffic flows across the network (e.g., Transmission Control Protocol [TCP] and Internet Protocol [IP], Open System Interconnection Model [OSI], Information Technology Infrastructure Library, current version [ITIL]).
- 150: Knowledge of what constitutes a network attack and the relationship to both threats and vulnerabilities.
- 214A: Skill in performing packet-level analysis.
- 214B: Knowledge of packet-level analysis using appropriate tools (e.g., Wireshark, tcpdump).
- 278: Knowledge of different types of network communication (e.g., LAN, WAN, MAN, WLAN, WWAN).

Centers of Academic Excellence Knowledge Units

- Basic Networking
- Network Forensics
- Network Technology and Protocols

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hp

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